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Nikhil Hegde

Software Engineer

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SUMMARY

Backend-dominant Full Stack Engineer with 2+ years of experience building scalable backend systems, AI/LLM-powered applications, and real-time distributed architectures. Experienced in designing high-throughput APIs, event-driven systems, and agentic AI workflows. Speaker and published researcher in Scopus Q1 journal.

SKILLS

Languages: Python, JavaScript, SQL, Java
Frameworks: FastAPI, Node.js, Express, Spring Boot, React
AI/LLM: Lang Chain, Lang Graph, RAG (Retrieval, Reranking, Context Optimization), Agents, Langfuse
Databases: PostgreSQL, MongoDB, Vector Databases (Pinecone, Qdrant)
Tools/Others: Docker, Git, AWS, Azure, Redis, WebSockets, Microservices, Async Programming

TECHNICAL TALK

MVCC in PostgreSQL: The Hidden Engine Behind Transactions

Aug - 2025
Hyderabad

- Delivered a deep technical talk on PostgreSQL internals to 200+ DBA professionals at Hyderabad PG Day 2025 covering MVCC, tuple visibility, transaction isolation, indexing internals, storage/page layout, and VACUUM mechanics.

EXPERIENCE

Senior Software Engineer

Stryv.ai

Mar 2026 – Present
Hyderabad, IN

- Recognized with the **Key Contributor Award** for impactful contributions to backend systems
- Architected and implemented an AI-powered semantic search and **dynamic website rendering** platform using RAG, enabling the UI to adapt intelligently based on user queries and significantly improving user experience.
- Built the end-to-end AI retrieval pipeline using Qdrant vector search, open-source embedding models, and reranking strategies, maintaining high retrieval accuracy while significantly **reducing inference costs** compared to cloud-based embeddings.
- Designed and developed an automated embeddings generation and ingestion pipeline integrating Sanity CMS and AWS S3, enabling scalable vector indexing, near real-time content synchronization, and efficient retrieval for AI-powered semantic search workflows.

Software Engineer

Stryv.ai

Aug 2024 – Mar 2026
Hyderabad, IN

- Optimized a backend service, achieving **78.6% reduction** in request latency under concurrent load, **63% reduction** for individual requests, and **4.7× throughput** improvement through architectural and performance optimizations.
- Designed and implemented a **supply-chain scoring engine** to compute order-level business metrics from complex business rules.
- Built a LangGraph-based **agentic AI chatbot** for a hospitality platform, automating customer support workflows with multi-step reasoning and tool-calling capabilities.

- Built a real-time user-facing **event streaming** system using Kafka and WebSockets to surface live process updates in the UI.
- **Reduced database load and improved API latency** through Redis caching, async design, and PostgreSQL query optimization.
- Migrated legacy .NET modules into FastAPI-based **microservices**, improving maintainability and system performance.

Software Engineer Intern

Ganglia Technologies Pvt Ltd

Jan 2024 – Jul 2024

Manipal, Karnataka

- Built a speech-to-text system for a healthcare application to **auto-populate clinical forms from voice input**.
- Designed and implemented **backend APIs** for a livestock marketplace platform, supporting product listings, search, filtering, and transaction workflows.
- Contributed to LLM-based retrieval workflows using **RAG** for structured data access.

EDUCATION

Manipal Institute of Technology

Master of Computer Applications (MCA) · 9.1/10 (Top 10% of the batch)

Aug 2022 – Aug 2024

Manipal, Karnataka

Karnataka University

Bachelor of Science (BSc) PMCs · 84%

Jun 2019 – Jul 2022

Dharwad, Karnataka

PROJECTS

- **Blog** : Write engineering blogs on **Medium** and **LinkedIn** covering backend and engineering topics, including articles like “**CORS Cannot Save Your Backend**”.
- **Mentorship** : Mentored developers at **React Hyderabad's Build with AI** event, guiding 150+ participants on **AI-assisted development** workflows and modern AI IDE adoption.
- Built multiple full-stack **personal projects**, including **Book A Ride** (Uber-like ride-booking platform), **InspireAI** (social platform with ML-based toxicity detection), and **LiveTrack** (real-time location tracking system similar to Swiggy/Zomato/WhatsApp Live Location).

CERTIFICATIONS

- **NodeJS Internals and Architecture By Hussein Nasser** Dec 2025
- **AWS Cloud Technical Essentials by Amazon Web Services (AWS)** Mar 2025
- **Complete Intro to React by Frontend Masters** Sep 2024
- **Machine Learning by Internshala Trainings** Aug 2023

RESEARCH PUBLICATIONS

Analyzing AI Regulation Through Literature and Current Trend

Mar 2025

- Published in Journal of Open Innovation (Scopus Q1, Top 1%)

Hospital-Acquired Infection Management with Artificial Intelligence: A Comprehensive Review

Sep 2024

- Presented at International Conference on IDEAAS 2024

Hybrid Approach - Diabetic Retinopathy Classification Through Activation Function Optimization

Dec 2024

- Presented at Springer Nature SIGMAA 2023